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Charging system and method of using same

Abstract

A charging system and a method of operating the same are provided. The charging system includes an electric machine which may be a wound or un-wound rotor type or a doubly fed induction motor (DFIM). A control system is coupled to the electric machine and a battery system. In the case of a wound rotor, the control system is coupled to stator windings and a rotor winding for controlling excitation of the stator windings and the rotor winding. The charging system is AC and DC compatible. In the case of an AC power source, the control system injects excitation into a rotor winding to induce a desired voltage in the stator, if the power supply voltage of the power supply is greater or smaller than the voltage of the battery system. Other modes of operation allowing for safe charging and discharging of a battery system are also described herein.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. provisional application 63/193,361, filed May 26, 2021, the contents of which are incorporated herein by reference in their entirety.

FIELD
(1) The specification relates generally to electric motors. This specification also relates to a charging system and method for using the same.
BACKGROUND OF THE DISCLOSURE
(2) The advent of electric vehicles has accelerated the adoption of new ways of consuming energy. With the increasing adoption of electric and hybrid vehicles, the demand for flexible energy management solutions is becoming increasingly apparent.
(3) With the increasing use of electric vehicles and the capability of charging battery packs in residential areas, there is a shift in the mix of energy consumption and storage. However, increasing the capacity of electric utility providers is an expensive endeavor that may prove uneconomical to serve the future needs of increased electric vehicle adoption and other means of electric power consumption.
(4) Existing approaches to mitigate this problem include utilizing solar panels at residential and other locations to feed a smart electric grid. Other approaches include the use of other renewable energy resources such as, wind, geothermal, hydropower and ocean resources, solid biomass, biogas and liquid biofuels to provide auxiliary power into the grid.
(5) However, improved management of other potential sources of energy, such as energy stored in vehicle battery systems are desired. For example, electric vehicle battery packs may be storing energy that may not be of immediate use to the vehicle owner that may be better utilized elsewhere.
(6) It is thus an object of the present disclosure to provide systems and methods to mitigate some of the aforementioned problems.
SUMMARY OF THE DISCLOSURE
(7) According to a part of the disclosure, there is provided a charging system comprising: an electric machine having at least three stator windings and at least one rotor winding; and a control system coupled to the at least three stator windings and the at least one rotor winding for controlling excitation of each of the at least three stator windings and the at least one rotor winding, wherein in at least one of a charging mode, a discharging mode: the control system determines if a power source voltage of a power source is outside of a voltage operating range of a battery system being charged or discharged; the control system applies an excitation voltage on the at least one rotor winding when the power source voltage is outside the voltage operating range of the battery system; and the control system terminates the excitation of each of the at least one rotor winding when the power source voltage is within the voltage operating range of the battery system.
(8) According to another part of the disclosure, there is provided a charging system, comprising: an electric machine having at least three stator windings and at least one rotor winding; and a control system coupled to the at least three stator windings and the at least one rotor winding for controlling excitation of each of the at least three stator windings and the at least one rotor winding, wherein in at least one of a charging mode or a discharging mode: the control system determines

the type of a power source. Upon determining that the type of the power source is DC: the control system bypasses the electric machine to charge the battery system from the power source, upon finding the power supply voltage Vs of the power supply is within a voltage operating range of a battery system; otherwise protects the battery system upon finding that Vs is greater than an upper limit of the voltage operating range. Otherwise upon determining that the type of the power source is AC: the control system connects a fast charger to the battery system; injects excitation into the at least one rotor winding to induce a voltage in the stator windings with 180 degree phase shift to the voltage of the power source, upon finding the power supply voltage Vs of the power supply is greater than the voltage of the battery system; otherwise terminating the excitation into one of the at least one rotor winding, upon finding the power supply voltage Vs of the power supply is not significantly lower than the voltage of the battery system; otherwise injecting excitation into one of the at least one rotor winding to induce an in-phase voltage.

(9) According to another part of the disclosure, there is provided a method of operating a charging system comprising an electric machine having at least three stator windings and at least one rotor winding; and a control system coupled to the at least three stator windings and the at least one rotor winding for controlling excitation of each of the at least three stator windings and the at least one rotor winding, the method comprising: in at least one of a charging mode or a discharging mode: determining the type of a power source; upon determining that the type of the power source is DC: bypassing the electric machine to charge the battery system from the power source, upon finding the power supply voltage Vs of the power supply is within a voltage operating range of a battery system; otherwise protecting the battery system upon finding that Vs is greater than an upper limit of the voltage operating range. Otherwise, upon determining that the type of the power source is AC: connecting a fast charger to the battery system; injecting excitation into the at least one rotor winding to induce a voltage in the stator windings with 180 degree phase shift to the voltage of the power source, upon finding the power supply voltage Vs of the power supply is greater than the voltage of the battery system; otherwise terminating the excitation into one of the at least one rotor winding, upon finding the power supply voltage Vs of the power supply is not significantly lower than the voltage of the battery system; otherwise injecting excitation into one of the at least one rotor winding to induce an in-phase voltage.

(10) This summary does not necessarily describe the entire scope of all aspects of the disclosure. Other technical advantages may become readily apparent to one of ordinary skill in the art after review of the following figures and description.

Description

BRIEF DESCRIPTIONS OF THE DRAWINGS

(1) For a better understanding of the embodiment(s) described herein and to show more clearly how the embodiment(s) may be carried into effect, reference will now be made, by way of example only, to the accompanying drawings in which:

(2) FIG. **1** is a schematic diagram of a charging system in accordance with an embodiment and its operating environment;

(3) FIG. **2** shows a schematic diagram of a converter of the charging system of FIG. **1**;

(4) FIGS. **3**A and **3**B collectively show a flow chart of a method of charging using the charging system of FIG. **1**;

(5) FIG. **3**C shows a flow chart of a magnetic coupling determination used by the charging system of FIG. **1**;

(6) FIG. **4**A shows the charging system of FIG. **1** in a motoring mode;

(7) FIG. **4**B shows the charging system of FIG. **1** in a configuration in charging/discharging mode supplied by DC power source with acceptable voltage;

(8) FIG. **4**C shows the charging system of FIG. **1** with a wound-rotor machine showing a configuration in charging/discharging mode when connected to a DC power source with low voltage;
(9) FIG. **4**D shows the charging system of FIG. **1** with a not-accessible-rotor machine showing a configuration in charging/discharging mode when connected to a DC power source with low voltage;
(10) FIG. **4**E shows the charging system of FIG. **1** showing a configuration in charging/discharging mode under protection;
(11) FIG. **4**F shows the charging system of FIG. **1** showing a configuration in charging/discharging mode when connected to an AC power source wherein the electric machine is bypassed;
(12) FIG. **5** shows waveforms of the charging system when excitation applied to decrease voltage at converter's terminals;
(13) FIG. **6** shows waveforms of the charging system when no excitation is applied;
(14) FIG. **7** shows waveforms of the charging system when excitation applied to increase voltage at the converter's terminals.
(15) Unless otherwise specifically noted, articles depicted in the drawings are not necessarily drawn to scale.

DETAILED DESCRIPTION

(16) For simplicity and clarity of illustration, where considered appropriate, reference numerals may be repeated among the Figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the embodiment or embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein may be practiced without these specific details. In other instances, well-known methods, procedures and components have not been described in detail so as not to obscure the embodiments described herein. It should be understood at the outset that, although exemplary embodiments are illustrated in the figures and described below, the principles of the present disclosure may be implemented using any number of techniques, whether currently known or not. The present disclosure should in no way be limited to the exemplary implementations and techniques illustrated in the drawings and described below.
(17) Various terms used throughout the present description may be read and understood as follows, unless the context indicates otherwise: “or” as used throughout is inclusive, as though written “and/or”; singular articles and pronouns as used throughout include their plural forms, and vice versa; similarly, gendered pronouns include their counterpart pronouns so that pronouns should not be understood as limiting anything described herein to use, implementation, performance, etc. by a single gender; “exemplary” should be understood as “illustrative” or “exemplifying” and not necessarily as “preferred” over other embodiments. Further definitions for terms may be set out herein; these may apply to prior and subsequent instances of those terms, as will be understood from a reading of the present description. It will also be noted that the use of the term “a” or “an” will be understood to denote “at least one” in all instances unless explicitly stated otherwise or unless it would be understood to be obvious that it must mean “one”.
(18) Modifications, additions, or omissions may be made to the systems, apparatuses, and methods described herein without departing from the scope of the disclosure. For example, the components of the systems and apparatuses may be integrated or separated. Moreover, the operations of the systems and apparatuses disclosed herein may be performed by more, fewer, or other components and the methods described may include more, fewer, or other steps. Additionally, steps may be performed in any suitable order. As used in this document, “each” refers to each member of a set or each member of a subset of a set.
(19) The present disclosure relates to a charging system adapted for charging and discharging to and from a battery pack. The act of discharging and charging may be facilitated by utilizing an existing electric machine and a motor controller, such as an inverter. As will be appreciated, the

demands for electric or hybrid vehicles are significant, and thus the power capacity of their batteries is large. In some circumstances, it can be desirable to employ the charge stored by the battery to power other external devices, such as, for example, to provide emergency power to critical electronic devices during a power outage.

(20) FIGS. **1** and **2** show a charging system **20** in accordance with an embodiment. The charging system **20** is integrated into the chassis of an electric vehicle **22**, and includes an electric machine **24**. The electric vehicle **22**, in effect, forms a housing for the charging system **20**. The electric machine **24** in this embodiment is a separately excited synchronous motor (SESM) that has two sets of windings. One set of stator windings **24** is on the stator, and the other set of windings **32** is on the rotor. In other embodiments, the electric machine **24** can be a wound rotor type such as separately excited synchronous motor or a doubly fed induction motor (DFIM). In yet other embodiments, the electric machine **24** can be non-wound rotor type such as permanent magnet, squirrel-cage induction, switched reluctance, and the like, and may include more than two sets of windings.

(21) An electrical current is passed through the rotor windings **32** using brushes, slip rings, or any other suitable manner to generate and control the field, commonly referred to as excitation. The stator windings **28** are shown in an open-end configuration, but can be set up in other configurations by a set of switches **57**, **58**. In the particular embodiment, there are three stator windings **28** and one rotor winding **32**.

(22) Electrical power is supplied to the electric machine **24** by a battery system **36**. The battery system **36** includes a set of one or more rechargeable battery packs **40** including cells connected in series or parallel providing a DC voltage. The battery packs **40** are managed by a battery management system (BMS) **44**. The BMS **44** monitors the mode of operation of the charging system **20**, operates to protect the battery packs **40** from operating outside their safe operating range, monitor and reporting the health of the battery packs **40**, and manages other parameters such as amount of allowed current and charging strategy (CCCV, etc.). The battery packs **40** and the BMS **44** of the battery system **36** may be any such components known in the art and adaptable for the charging system **20**.

(23) The charging system **20** enables alternation between operating modes; in particular, a motoring mode, a charging mode, and a discharging mode. In the motoring mode, power is provided from the battery system **36** to the electric machine **24**, feeding both the stator windings **28** and the rotor windings **32** in accordance with a desired scheme for causing the rotor to rotate, thereby transferring power to a transmission attached to the rotor. In a charging mode, power is received from an external power source, such as a power grid, to charge the battery packs **40** of the battery system **36**. The external power source is bidirectional, meaning that it can both provide and receive power. The power source **68** can include, for example, a power grid, a fast charger, and a wireless power transceiver. The power received from or transmitted to the power source **68** can be alternating current (AC) or direct current (DC) provided by a utility grid, a fast charger, a wireless charger, etc. In a discharging mode, power is transferred from the battery packs **40** to a power load. The transferred power can be AC or DC, depending on the needs of the power load.

(24) In order to control operation of the charging system **20** to alternate between modes of operation and to accommodate variances in the voltage of the power received during charging or to be supplied during discharging, a control system **48** communicates with the battery system **36** to synchronize the operating mode, and it controls operation of the electric machine **24**, an input/output (I/O) selector **52**, and a switch box **56**. The I/O selector **52** is interposed between the electric machine **24** and the control system **48** and selects whether or not to involve the motor during the charging/discharging procedure. This is achieved by the I/O selector **52** making a connection between the battery system **36** and the electric machine **24** or a power source or load, such as a power grid **50**, via two sets of switches. A set of battery-motor switches **53** connect or disconnect the battery system **36** to the electric machine **24** via the converter **72**, and a set of

battery-power source switches **54** connect or disconnect the battery system **36** to a power source **68** that includes the power grid **50**. The I/O selector **52** allows bypass of the motor and use of additional inductors to perform the charging or discharging functionality. This is needed in a case that the motor is not of a wound-rotor type (e.g., permanent magnet, squirrel-cage induction, switched reluctance, etc.). It should be noted that a charging system will need either the I/O selector **52** or the switch box **56** according to design specifics. The power grid **50** is a grid of utility or any other type that provides energy and/or receives energy back. The grid type can be AC or DC, with at least 2 phases, or 1 phase and grid neutral, or DC. The switch box **56** is interposed between the electric machine **24** and the power grid **50**, and controls coupling of the motor windings **28** to each other via a set of stator winding connector switches **57** or to the power grid **50** via a set of motor-power source switches **58**. There is a set of switches **59** between the power source **68** and the battery pack **36** which enable to bypass the electric machine **24** and the converter **72** when the power source **68** is DC with an amplitude that is suitable for charging. The switches **53**, **54**, **57**, **58**, and **59** can be power switches, relays, contactors, or any other suitable type of switch. In the case that the number of rotor windings of the motor is less than three, the motor-power source switches **58** can have the capability of independent-pole control.

(25) While the charging system **20** is shown having a separately excited synchronous motor (SESM) as electric machine **24**, in other embodiments, it can alternatively include a doubly fed induction motor (DFIM). Further, the electric motor can also be a motor with non-wound rotor, but, in these scenarios, the motor is not involved during the charging/discharging procedure.

(26) Connection to the power grid **50** can be made through either a wireless or wired interface. A wireless power transceiver **60** of the charging system **20** within the electric vehicle **22** is wirelessly coupled to a wireless power transceiver **64** positioned outside the electric vehicle **22** connected to the power grid **50**. A wired connection is also can be made through a line filter **66**.

(27) The control system **48** includes a converter **72**, shown in greater detail in FIG. **2**, for converting direct current to alternating current and vice versa. The illustrated converter **72** is a 2-level, 3-phase converter. In other embodiments, the converter can be any suitable type of multi-level, multi-phase converter. The converter **72** controls the flow of electricity from the battery system **36** to the electric machine **24** or to the power grid **50** or from the power grid **50** to the battery system **36**. The converter **72** includes a split dc-link. Accessible DC-link middle point **73** gives the converter **72** the capability of independent pole control, which makes it suitable for both single and 3-phase systems. To provide independent pole control the mid-point of the dc-link **73** may be connected to the power source neutral.

(28) A field control module **76** of the control system **48** controls a flow of electricity from the battery system **36** to the rotor winding(s) **32**. The field control is realized using an H-bridge or a similar scheme.

(29) A charge/discharge control module **80** of the control system **48** is in communication with the switch box **56** that controls coupling of the motor's stator windings **28** either to each other or to the power side which can be power grid **50** or fast charger **45**.

(30) A monitoring and communication module **84** of the control system **48** is in communication with the power source **68** to decide the mode and manage the rate of charging or discharging based on the information it gets from an energy management system **88** and the battery management system **44**. In addition, the monitoring and communication module **84** is in communication with the energy management system **88** for managing, provisioning, and billing. Support is provisioned in the monitoring and communication module **84** for proprietary or external management systems, cloud or other type. Feedback is provided about the amount of energy exchanged in charging or discharging mode, as well as the health of the battery system **36** shared via the BMS **44** communication protocol.

(31) FIGS. **3**A and **3**B show the general method **100** of determining the configuration and operation of the charging system **20**. The method **100** commences with the determination of whether the

charging system **20** should be operated in the charging mode or the discharging mode (**105**), based on the information on utility and battery pack. The monitoring and communication module **84** of the control module **48** determines whether the charging system **20** is to be in a charging, discharging, or monitoring state. To make this decision, the monitoring and communication module **84** communicates with the BMS **44** to determine if the battery system **36** is able to charge, discharge, both, or none. Then the monitoring and communication module **84** communicates with the energy management system **88** to determine the user-selected mode of operation. The monitoring and communication module **84** can maintain a user account, an identifier of a user account managed or accessible to the energy management system **88**, or an identifier that is associated with a user account managed or accessible to the energy management system **88**. The user account includes a user-specified configuration as to how the charging system **20** is to behave under various circumstances. For example, the user account can indicate that discharging is permissible during a time range, and a minimum level of battery system capacity to be maintained. If the energy management system **88** is unavailable or no user account is specified, a default state (charging or discharging) is enabled, depending on the user configuration.
(32) If the charging system **20** is not to be configured and operated in the charging mode or the discharging mode, the charging system **20** is configured in the motoring mode (**106**).
(33) FIG. **4**A shows the charging system **20** configured in the motoring mode. The battery-motor switches **53** of the I/O selector **52** are closed and the battery-power source switches **54** are opened to connect the battery system **36** to the electric machine **24** via the converter **72**. In addition, the stator winding connector switches **57** are closed and the motor-power source switches **58** and fast charger switches **59** are opened, thereby isolating the electric machine **24** from the power source **68** and connecting the stator windings **28** to each other in the appropriate configuration (e.g., Wye, etc.). As a result, the electric machine **24** is coupled to the battery system **36** and decoupled from the power grid **50**. In this mode, the electric machine **24** can act as a motor to drive the electric vehicle **22**. The control system **48** implements a motor control strategy, such as FOC. If the electric machine is of a wound-rotor type, then while in motoring mode, the field control module **76** can affect a field excitation in the rotor winding(s) **32** of a constant current, unity flux control, unity power factor, or any other control type to allow operation of the motor.
(34) Referring again to FIGS. **1**, **3**A, and **3**B, the control module **48** determines a configuration and operation mode matching the detected power source **68**, its characteristics (that is, its maximum voltage amplitude and type), and the condition of the battery pack **40**. If the charging system **20** is charging or discharging, the power characteristics of the power source are determined (**110**). To properly control the charging system **20**, the control system **48** needs to know characteristics of the voltage of the power source **68**. In particular, the monitoring and protection module **70** measures power source **68** voltages and communicates with control system **48**. The control system **48** finds the power source angle and frequency using an identification method algorithm e.g. a Phase Lock Loop (PLL). The power source type (AC or DC) is detected based on detected frequency, and voltage amplitude is found using measured instantaneous values by monitoring and protection module **70** and detected voltage angle.
(35) Using the characteristics of the power source **68** determined at **110**, it is determined if the power source **68** is alternating current (AC) or direct current (DC) (**120**). If the power source **68** is determined to be DC, it is determined if the voltage V.sub.s of the power source **68** is within acceptable limits of the battery system **36** (**130**). The BMS **44** maintains specifications for safe operation of the battery system **36**, including a voltage range that can be received. In particular, the control system **48** communicates with the monitoring and protection module **70** to compare the voltages of the battery system **36** and the power source **68**. If the voltage V.sub.s of the power source **68** is within the acceptable limits, the converter **72** and the electric machine **24** are bypassed (**140**). The control system **48** directs the I/O selector **52** to open the battery-motor switches **53** and the battery-power supply switches **54**, and the control module **80** directs the switch box **56** to open

the motor-power source switches **58** to decouple the electric machine **24** from the power source **68** and to close the fast-charger switches **59**, thereby decoupling the electric machine **24** and the converter **72** from the battery system **36** and connecting the DC power source **68** directly to the battery system **36**.

(36) FIG. **4**B shows the fast-charger switch **59** in a closed position so that charging and discharging of the battery system **36** is performed directly via the fast charger **45** of the power source **68**. Both the converter **72** and the electric machine **24** are not used in this configuration. Any suitable charging algorithm can be employed, such as CCCV.

(37) Referring again to FIGS. **1**, **3**A and **3**B, if, instead, it is determined at **130** that the voltage V.sub.s of the power source **36** is not within the acceptable limits, it is determined if the voltage V.sub.s is greater than an upper limit of the battery system **36** (**150**). If the voltage V.sub.s is less than the lower limit, it is determined if the electric machine **24** includes a wound rotor (**160**). If the electric machine includes a wound rotor, excitation of the rotor winding(s) **32** is terminated and the electric machine stator winding(s) is/are used as the boost inductors (**170**). The control system **48** puts two of the stator windings **28** of the electric machine **24** between the power source **68** and the converter **72** to use the converter **72** to boost the voltage to an acceptable level. To do so, the control system **48** directs the I/O selector **52** to open the battery-power source switches **54** and close the battery-motor switches **53**, and it directs the charge/discharge control module **80**, to open the stator winding connector switches **57**, and the fast-charger switches **59** and close motor-power source switches **58**.

(38) FIG. **4**C shows the configuration of the charging system **20** in this condition. Each stator winding **28** is connected separately to the power source **68** and there is no connection between the stator windings **28**. The electric machine **24** is interposed between the power source **68** and the converter **72**. The control system **48** implements a control strategy via the converter **72**. In charging mode, the stator windings **28** are used as inductors, and the converter **72** acts as a DC/DC converter. The control system **48** verifies that the charging system **20** is stationary, as it is not possible to use the electric machine **24** as a motor during charging or discharging. In discharging mode, power flows back to the power source **68**. The motor windings **28** are used as inductors, and the converter **72** acts as a DC/DC converter in buck mode.

(39) Returning again to FIGS. **1**, **3**A, and **3**B, if the electric machine **24** is not a wounded-rotor type, the control system **48** inserts inductors between the converter **72** and the power source **68** (**180**). The control system **48** closes the battery-power source switches **54** and opens the battery-motor switches **53**, the motor-power source switches **58**, and the fast-charger switches **59**.

(40) FIG. **4**D shows system configuration in this condition, wherein the control system **48** directs the I/O selector **52** to open the battery-motor switches **53** and close the battery-power source switches **54**, thereby decoupling the electric machine **24** from the battery system **36**. The charge/discharge control module **80** directs the switch box **56** to open the motor-power source switches **58** to decouple the electric machine **24** from the power grid **50**.

(41) Returning to FIGS. **1**, **3**A, and **3**B, if, instead, it is determined that the voltage V.sub.s is greater than the upper limit at **150**, then the control system **48** protects the converter **72** of the battery system **36** against overvoltage (**190**). To do so, the control system **48** directs the battery-power source switches **54**, the motor-power source switches **58**, and the fast-charger switches **59** to open.

(42) FIG. **4**E shows system configuration in this condition.

(43) Returning again to FIGS. **1**, **3**A, and **3**B, if it is determined that the power source **68** is AC at **120**, the control system **48** directs the charge/discharge control module **80** to open the fast-charger switches **59** (**220**). Then it is determined if the electric machine type includes rotor windings (**230**). If the electric machine does not include accessible rotor windings, it is determined if the voltage V.sub.s of the power source is greater than the acceptable operation voltage V.sub.b of the battery system **36** (**240**). If the voltage V.sub.s of the power source is determined not to be greater than the

acceptable operation voltage V.sub.b of the battery system **36** at **240**, then the electric machine is bypassed (**250**). The control system **48** directs the I/O selector **52** to close the battery-power source switches **54** and open battery-motor switches **53**. The charge/discharge control module **80** directs the switch box **56** to open switch **58** to decouple the electric machine **24** from the power source **68**.

(44) FIG. **4**F shows the configuration of the charging system **20** in these circumstances. In this condition, the converter **72** along with the provided inductors act as a boost converter to charge/discharge the battery system **36**.

(45) Returning again to FIGS. **1**, **3**A, and **3**B, in the case that electric machine does not include accessible rotor winding, if the voltage V.sub.s of the AC power source is determined to be greater than the voltage V.sub.b of the battery system **36** at **240**, then the control system **48** protects the battery system **36** and the converter **72** against overvoltage (**260**). The control system **48** directs the I/O selector **52** to open switch **54**. To do so, the control system **48** directs the battery-power source switches **54**, the motor-power source switches **58** to open, therefore, decouples the electric machine **24** and converter **72** from the power source

(46) FIG. **4**E shows the configuration of the charging system **20** in this condition.

(47) Returning again to FIGS. **1**, **3**A, and **3**B, if, instead, the electric machine type is a wound-rotor type (that is, any electric machine with accessible windings on the rotor, such as a separate-excited synchronous machine (SESM) or a doubly fed induction machine (DFIM)), it is determined if the voltage V.sub.s of the power source **68** is greater than the voltage V.sub.b of the battery system **36** (**310**). If the voltage V.sub.s of the power source **68** is greater than the voltage V.sub.b of the battery system **36**, the control system **48** synchronizes the excitation current which is fed into the rotor winding(s) **32** with the voltage V.sub.s of the power source **68** in such a way that it induces voltages with a 180-degree phase shift in respect to the voltages of the power source **68** (**340**) into stator windings. This synchronization can be done by conventional PLL or any other suitable method. In the case that the electric machine **24** is a SESM (**320**), the stator winding with the highest magnetic coupling to rotor is found (**330**), shown in FIG. **3**C at steps **301**, **302** and **303**, and synchronization and configuration is done accordingly. After synchronization, the control system **48** configures the charging system **20**. The battery-motor switches **53** of the I/O selector **52** are closed and the battery-power source switches **54** are opened. Further, the stator winding connector switches **57** are opened and the motor-power source switches **58** are closed. FIG. **4**C shows this configuration. The motor-power source switches **58** have independent pole operation. In the case where the electric machine **24** is a SESM, the pole connected to the stator winding with the highest magnetic coupling to the rotor is connected and two other poles remain open. Each of the stator windings **28** is connected separately to the power source **68** and there is no connection between the stator windings **28**. This enables the voltage amplitude at the terminals of the converter **72** to be lowered and to keep the converter **72** at a boost region where it has controllability. FIG. **5** shows the voltage and current waveforms in this working condition.

(48) If the voltage V.sub.s of the power source **68** is determined to be lower than or equal to the voltage V.sub.b of the battery system **36** at **310**, it is determined if the voltage V.sub.s of the power source **68** is significantly lower than the voltage V.sub.b of the battery system **36** (**360**) i.e. when the control system **48** reaches the converter **72** boost or buck ratio limit, without reaching the required voltage to charge or discharge the battery system **36**, respectively. If the voltage V.sub.s of the power source **68** is not significantly lower than the voltage V.sub.b of the battery system **36**, it is deemed that the V.sub.s of the power source **68** is at an appropriate level for the converter **72** and battery system **36**. Therefore, excitation of the rotor winding(s) is terminated and no current is injected into the rotor winding(s) (**370**). The charging system **20** is then reconfigured at **350** as FIG. **4**C. FIG. **6** shows the voltage and current waveforms when the charging system **20** is configured as shown in FIG. **4**C.

(49) If the voltage V.sub.s of the power source **68** is significantly lower than the voltage V.sub.b of the battery system **36**, the electric machine type is determined (**380**). If the electric machine type is

a separately excited synchronous motor, the stator winding with the highest magnetic coupling to rotor is found (**385**) also shown in FIG. **3**C at steps **301**, **302** and **303**, and synchronization is done accordingly. After synchronization, the control system **48** configures the charging system **20** as shown in FIG. **4**C. The battery-motor switches **53** of the I/O selector **52** are closed and the battery-power source switches **54** are opened. Further, the stator winding connector switches **57** are opened and the motor-power source switches **58** are closed. The motor-power source switches **58** have independent pole operation. In the case where the electric machine **24** is a SESM, the pole connected to the stator winding with the highest magnetic coupling to the rotor is connected and two other poles remain open. Each of the stator windings **28** is connected separately to the power source **68** and there is no connection between the stator windings **28**. This enables the voltage amplitude at the terminals of the converter **72** to be increased and to keep the converter **72** at a boost region with proper boost ratio. Control system **48** synchronizes the excitation current which is fed into the rotor winding(s) **32** with the voltage V.sub.s of the power source **68** in such a way that it induces voltages in-phase with the voltages of the power source **68** (**390**) into stator windings. By doing so, the amplitude of the voltage at the terminal of the converter **72** is raised to keep the boost ratio at an acceptable value. In particular, where the voltage level being received is insufficient to charge the battery system **36** in a proper manner, it is increased to at least to the minimum level required for this purpose, as shown in FIG. **7**. The charging system **20** is then reconfigured at **350**

(50) As will be appreciated, the charging system **20** does not require external transformers such as those implemented at charging stations, DC/DC converters, or AC/DC converters such as those on-board in the case of mobile applications (e.g., an electric vehicle).

(51) Thus, by using the I/O selector **52** and the switch box **56**, the electric machine **24** can be selectively used to condition incoming or outgoing power based on the requirements of both the battery system **36** and the power source **68**.

(52) The system described herein is designed to be compatible with any grid system operating on DC or AC voltage and power, and with any electric machine.

(53) While, in the above embodiment, the charging system is shown as a three-phase system, it will be appreciated that the charging system can be configured for a system with any number of phases.

(54) In other embodiments, the charging system as described herein can be used for other types of applications, such as other types of vehicles or devices. In one particular example, an electric vehicle parked in a parking garage can be used to provide emergency loads during an outage and for stabilizing a power grid in emergency situations or participate in a vehicle-to-everything (V2X) scenario in a vehicular communication system.

(55) It will be appreciated that the rotor can have more than one rotor winding and that a subset of the rotor windings can be controlled to provide the benefits described hereinabove.

(56) Further, while particular reference has been made to power grids, other types of power sources or power loads can be used with the charging system.

(57) Although specific advantages have been enumerated above, various embodiments may include some, none, or all of the enumerated advantages.

(58) Persons skilled in the art will appreciate that there are yet more alternative implementations and modifications possible, and that the above examples are only illustrations of one or more implementations. The scope, therefore, is only to be limited by the claims appended hereto and any amendments made thereto.

LIST OF REFERENCE NUMERALS

(59) **20** charging system **24** electric machine **28** motor windings **32** rotor winding(s) **36** battery system **40** battery packs **44** BMS **45** fast charger **48** control system **50** power grid **52** I/O selector **53** battery-motor switches **54** battery-power source switches **56** switch box **57** stator winding connector switches **58** motor-power source switches **59** fast-charger switches **60** wireless power transceiver **64** wireless power transceiver **66** line filter **68** power source **70** monitoring and

protection module **72** converter **73** mid-point of the dc-link **74** converter bypass switches **76** field control module **80** charge/discharge control module **84** monitoring and communication module **88** energy management system **100** method **105** charging or discharging? **106** operate in motoring mode **110** identify power characteristics **120** is power source AC? **130** is V.sub.s within acceptable limits? **140** bypass converter and electric machine **150** is V.sub.s greater than upper limit? **160** Is electric machine wound rotor? **170** terminate excitation and use electric machine as boost inductors **180** bypass electric machine **190** protect battery pack converter **220** open fast-charger switch **230** is EM type wound rotor? **240** V.sub.s>V.sub.b? **250** bypass electric machine **260** protect battery system and converter **310** V.sub.s>V.sub.b? **320** electric machine type? **330** find stator winding high coupling to rotor **340** inject sync excitation to induce voltage with 180 degree phase **350** reconfigure system **360** V.sub.s<<V.sub.b? **370** terminate excitation **380** electric machine type? **385** find stator winding high coupling to rotor **390** inject sync excitation to induce in-phase voltage **301** detect and measure rotor angle **302** detect coupling to rotor **303** return phase with highest coupling to the rotor

Claims

1. A method of operating a charging system, the charging system comprising: (i) an electric machine comprising at least three stator windings and at least one rotor winding; and (ii) a control system comprising a converter coupled to the at least three stator windings and the at least one rotor winding for controlling excitation of each of the at least three stator windings and the at least one rotor winding; the method comprising: (a) Operating in at least one of a charging mode or a discharging mode; and (b) determining if a type of a power source is DC or AC; and (i) upon determining that the type of the power source is DC, then performing one of the following steps: (A) bypassing the electric machine to charge a battery system from the power source, upon finding the power supply voltage Vs of the power supply is within a voltage operating range of the battery system; (B) using the stator windings as inductors wherein the converter acts as DC/DC in buck or boost mode, in said discharging or charging mode, respectively; (C) protecting the battery system upon finding that Vs is greater than an upper limit of the voltage operating range; and (ii) upon determining that the type of the power source is AC, then performing one of the following steps: (A) injecting excitation into the at least one rotor winding to induce a voltage into the at least one of the stator windings with 180 degree phase shift to the voltage of the power source, upon finding the power supply voltage Vs of the power supply is greater than the voltage of the battery system; (B) terminating the excitation into one of the at least one rotor winding, upon finding the power supply voltage Vs of the power supply is not significantly lower than the voltage of the battery system; (C) injecting excitation into the at least one rotor winding to induce a voltage into the at least one of the stator windings in-phase with the voltage of the power source, upon finding the power supply voltage Vs of the power supply is smaller than an acceptable voltage of the battery system.
2. The method of claim 1, wherein the electric machine is a separately excited synchronous machine (SESM).
3. The method of claim 2, further comprising finding the one of the stator windings having the highest magnetic coupling to the rotor prior to injecting excitation into one of the at least one rotor winding.
4. The method of claim 1, wherein the connecting the fast charger to the battery system comprises, closing the set of fast-charger switches.
5. The method of claim 1, wherein the electric machine is a doubly fed induction machine (DFIM).
6. The method of claim 1, wherein the converter connects the battery system to the electric machine.
7. The method of claim 1, wherein upon said protecting the battery system, the set of fast-charger

switches are opened.
8. The method of claim 1, wherein the power source comprises one or more of: a fast charger, a wireless power transceiver and a power grid.
9. The method of claim 1, wherein the battery comprises a plurality of battery packs and a battery management system.
10. The method of claim 1, wherein the charging system further comprises a monitoring and protection module in communication with a set of motor-power source switches interposed between the electric machine and the monitoring and protection module.
11. The method of claim 1, wherein the charging system further comprises a set of battery-motor switches, the method further comprising using the set of battery-motor switches to disconnect the battery system from the electric machine.
12. A charging system comprising: (a) an electric machine having at least three stator windings and at least one rotor winding; and (b) a control system coupled to the at least three stator windings and the at least one rotor winding for controlling excitation of each of the at least three stator windings and the at least one rotor winding; wherein: (a) the charging system is configured to operate in at least one of a charging mode or a discharging mode; and (b) the control system is configured to determine whether a type of a power source is AC or DC; and (c) the charging system is configured to perform any one of the following steps: (i) if the power source is DC, then (A) charge a battery system from the power source, upon finding the power supply voltage Vs of the power supply is within a voltage operating range of the battery system, or (B) use the electric machine's windings as inductors and the converter acts as DC/DC in buck or boost mode, in discharging or charging mode, respectively (C) protect the battery system upon finding that Vs is greater than an upper limit of the voltage operating range; and (ii) if the power source is AC, then: (A) inject excitation into the at least one rotor winding to induce a voltage in the at least one of the stator windings with 180 degree phase shift to the voltage of the power source, upon finding the power supply voltage Vs of the power supply is greater than the voltage of the battery system; (B) terminate the excitation into one of the at least one rotor winding, upon finding the power supply voltage Vs of the power supply is not significantly lower than the voltage of the battery system; or (C) inject excitation into the at least one rotor winding to induce a voltage into the at least one of the stator windings in-phase with the voltage of the power source, upon finding the power supply voltage Vs of the power supply is smaller than an acceptable voltage of the battery system.
13. The charging system of claim 12, wherein the electric machine is a separately excited synchronous machine (SESM).
14. The charging system of claim 13, prior to injecting excitation into one of the at least one rotor winding, the control system finds the one of the stator windings having the highest magnetic coupling to the rotor.
15. The charging system of claim 12, further comprising a set of fast-charger switches between the power source and the battery system, wherein the connecting the fast charger to the battery system comprises, closing the set of fast-charger switches.
16. The charging system of claim 12, wherein the electric machine is a doubly fed induction machine (DFIM).
17. The charging system of claim 12, wherein the control system comprises: a converter for connecting the battery system to the electric machine.
18. The charging system of claim 17, wherein the converter comprises a split dc-link.
19. The charging system of claim 18, wherein the power source type is AC and a midpoint of the split dc-link is electrically connected to a neutral line of the power source.
20. The charging system of claim 19, further comprising motor-power source switches enabling independent pole operation when desired.
21. The charging system of claim 15 further comprising a set of battery-motor switches and a set of battery-power source switches, wherein upon said protecting the battery system, the set of fast-

charger switches, the set of battery-motor switches, and the set of battery-power source switches are opened.

22. The charging system of claim 12, wherein the power source comprises one or more of: a fast charger, a wireless power transceiver and a power grid.

23. The charging system of claim 12, wherein the battery comprises a plurality of battery packs and a battery management system.

24. The charging system of claim 12, further comprising a monitoring and protection module in communication with a set of motor-power source switches interposed between the electric machine and the monitoring and protection module.

25. The charging system of claim 12, further comprising a set of battery-motor switches that connect or disconnect the battery system to the electric machine.

26. A charging system comprising: (a) a non-wound rotor type electric machine; and (b) a control system coupled to the electric machine wherein upon determining the voltage of a power source, charging or discharging a battery system can be enabled by bypassing the non-wound rotor type electric machine.